Human Activity Recognition

USING SMARTPHONE SENSOR DATA

A Comparative Machine Learning Study

COMP4030 – Data Science with Machine Learning | Spring 2026
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Research Overview

Why This Study?

HAR from smartphone sensors underpins fall detection, rehab monitoring, and fitness tracking.

Most studies compare only 3–5 classifiers with no statistical testing gaps may be noise.

Standard random splits let training/test subjects overlap, inflating reported accuracy for real deployments.

This study addresses all three gaps with 10 classifiers, formal Wilcoxon testing, and subject-independent LOGO evaluation.

Research Questions

RQ1

Dimensionality

Can PCA-reduced features maintain accuracy vs. all 561 features and where does performance plateau?

RQ2

Model Comparison

Do ensemble methods significantly outperform linear baselines, confirmed by Wilcoxon signed-rank testing?

RQ3

Generalisation

Does Leave-One-Subject-Out (LOGO) evaluation reveal meaningful performance degradation vs. random splitting?

Dataset & Methodology

WALKING | WALK UP | WALK DN | SITTING | STANDING | LAYING

30 healthy adults (ages 19–48) | Samsung Galaxy S II on waist | Triaxial accel. + gyro @ 50 Hz | 2.56-sec sliding windows → 561 features per window

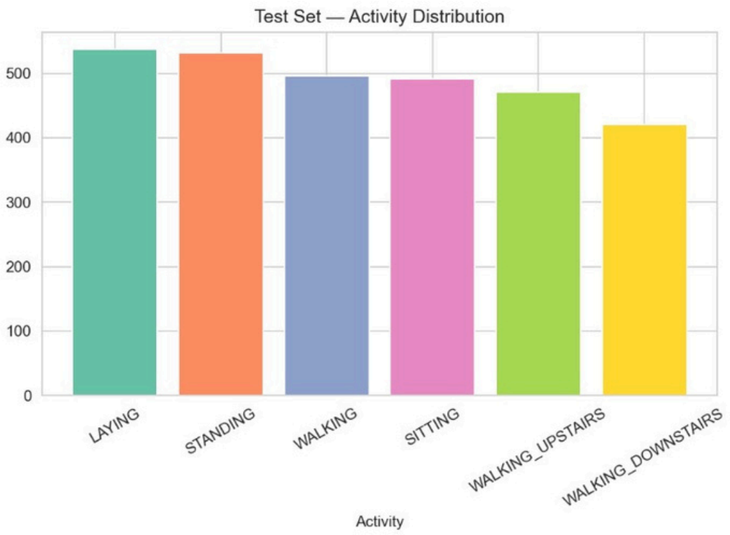
Official Split

Partition	Subjects	Samples
Training	21	7,352
Test	9	2,947
Total	30	10,299

Pipeline

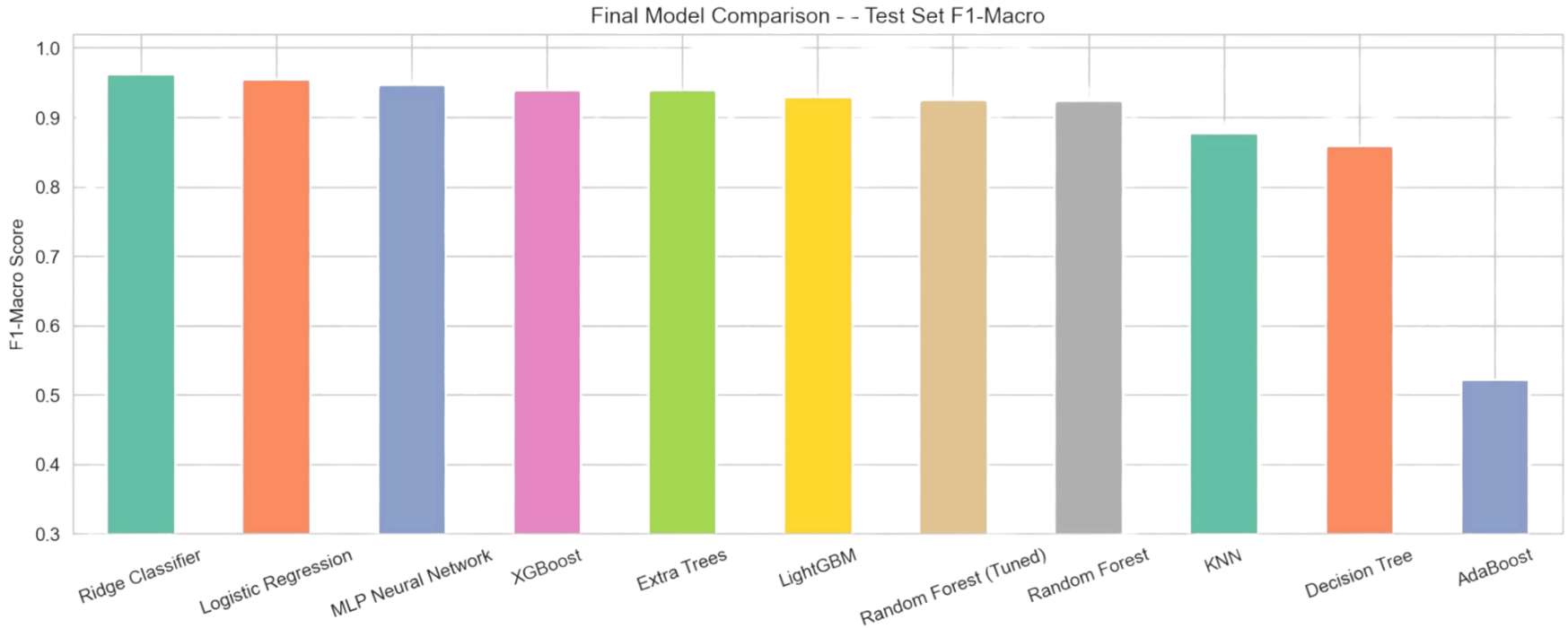
1. Data Loading & Preprocessing → 2. Exploratory Data Analysis (EDA) → 3. Dimensionality Reduction using PCA → 4. Model Training & Cross-Validation → 5. Statistical Comparison of Models → 6. Subject-Independent Evaluation (LOGO)

$$F1\text{-}macro = (1/C) \times \Sigma\, F1_c \quad | \quad F1_c = 2\cdot(P_c \cdot R_c) / (P_c + R_c) \quad | \quad C = 6 \text{ activity classes}$$



- Smartphone accelerometer and gyroscope data were used to classify six human activities.
- A machine learning pipeline with PCA, cross-validation and macro F1-score was used for fair model comparison.

Cross-Validation Results — 10-Fold F1-Macro



0.9933
LightGBM
Best in CV

0.9871
Logistic Reg.
Linear baseline

0.3855
AdaBoost
Decision stumps fail

10 Classifiers — Theoretical Justification

Linear

LR, Ridge:

High-dim scaled space → linear boundaries; t-SNE confirms separability

Bagging

RF, Extra Trees:

Random feature subsampling handles multicollinearity in 561 features

Boosting

XGB, LGBM, AdaBoost:

State-of-the-art tabular; tests iterative correction vs. bagging

Other

KNN, MLP:

Distance-based baseline + non-linear upper bound

Test Set Results — The Surprise: Ridge Beats LightGBM

Classifier	CV F1-Macro	Test F1-Macro	Gap
Ridge Classifier ★	0.9866	**0.9624**	-0.0242
Extra Trees	0.9907	0.9611	-0.0296
Logistic Regression	0.9866	0.9591	-0.0275
MLP Neural Network	0.9902	0.9566	-0.0336
Random Forest	0.9822	0.9530	-0.0292
XGBoost	0.9933	0.9389	**-0.0544** ⚠
LightGBM	0.9938	0.9287	**-0.0651** ⚠
AdaBoost	0.3855	0.5218	*+0.1363*

Why ?

Ridge beats LightGBM on the test set despite a lower CV score.

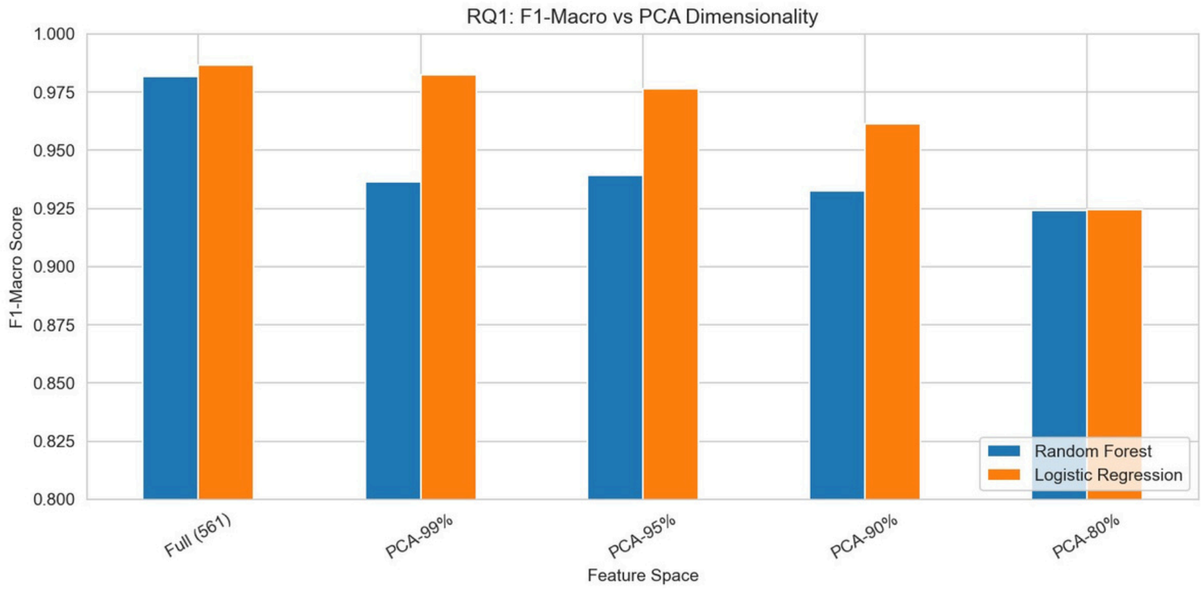
During 10-fold CV, all folds share the same 21 training subjects — boosting methods memorise subject-specific movement patterns.

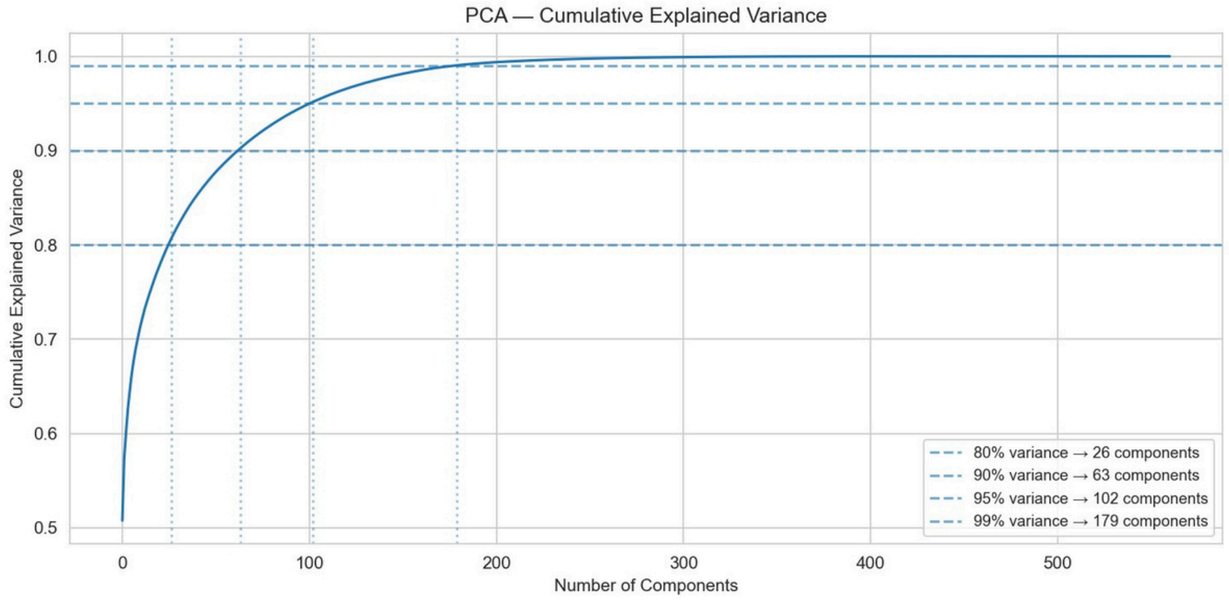
On 9 completely new test subjects, this memorised knowledge becomes a liability.

Ridge's L2 regularisation prevents memorisation — its inflexibility becomes a strength on unseen individuals.

Surpasses Anguita et al. SVM baseline by +7.2pp and matches Ordóñez CNN-LSTM without raw signal access.

RQ1 — PCA: Does Dimensionality Reduction Hurt?





RQ1 Answer

✓ Full 561 ≈ PCA-95% (102 components) — near-zero accuracy loss

✓ 95% variance captured by only 18% of features — massive redundancy confirmed

✓ Performance degrades measurably only below 80% variance threshold

→ PCA-95% is the optimal compression point

! Real-world impact: faster inference on smartwatches with limited RAM

RQ2 — Statistical Testing | RQ3 — Generalisation Gap

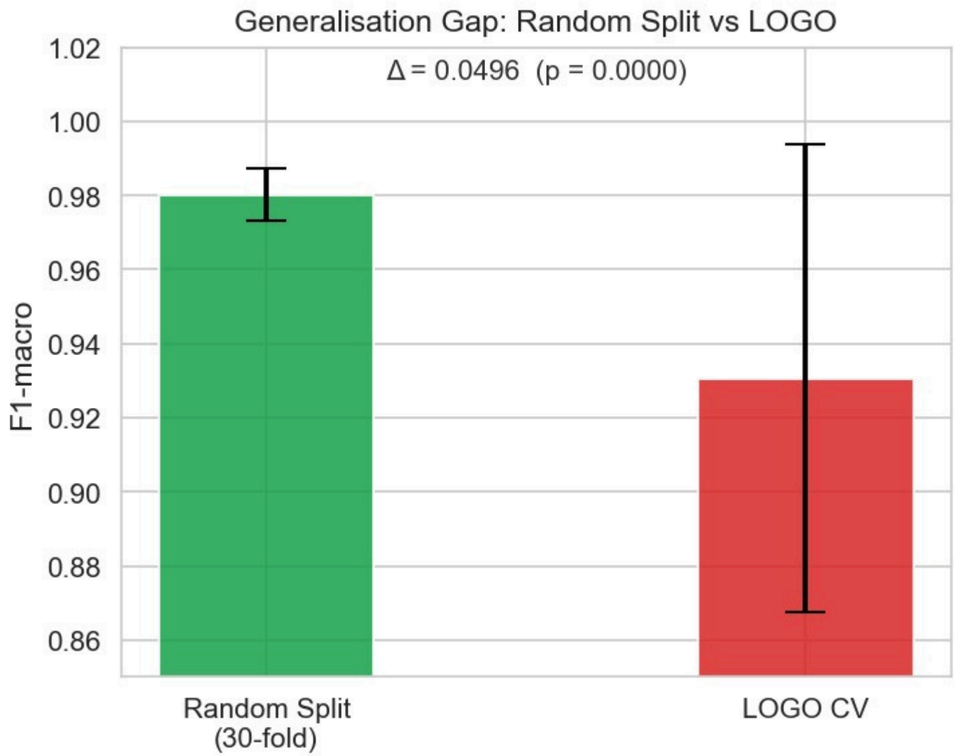
RQ2 — Wilcoxon Signed-Rank Tests

Friedman Omnibus: $\chi^2 = 80.929$, $p = 0.0000$ → Significant differences confirmed

Comparison	p-value	Result
LR vs Extra Trees	p = 1.000	**LR = ET**
Ridge vs RF	p = 0.432	**Ridge = RF**
XGBoost vs LightGBM	p = 0.922	Equivalent
LightGBM vs LR	p = 0.002	**LGBM > LR ✓**
AdaBoost vs all	p = 0.002	**Ada worst ✗**

RQ2 Answer: Ensembles do NOT universally beat linear models. Only gradient boosting shows consistent superiority in CV — and this advantage collapses on the test set.

RQ3 — LOGO vs Random Split



RQ3 Answer: Standard eval inflates performance by ~5%. LOGO's higher variance (σ=0.063) shows some subjects are harder to generalise to.

Conclusion

1. Ridge wins at test time

F1-macro = 0.9624, outperforming all gradient boosting despite lower CV — regularisation beats memorisation. Surpasses Anguita SVM baseline by +7.2pp.

2. PCA: 18% of features is enough (RQ1)

95% variance retained by 102 of 561 components with near-zero accuracy loss. Supports efficient deployment on resource-constrained embedded systems.

3. Ensembles ≠ always better

Wilcoxon: LR is statistically equivalent to Extra Trees ($p=1.000$); Ridge equals Random Forest ($p=0.432$). Only gradient boosting shows consistent CV superiority — which collapses at test time.

4. 5% generalisation gap (RQ3)

LOGO exposes $\Delta=0.0496$ vs random splitting ($p<0.0001$). Standard random-split evaluation inflates reported performance — clinically significant in healthcare deployments.